

Introductory Real Analysis: Square Roots

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1 What is $\sqrt{2}$?

We understand that it should exist, since in a right triangle with legs $a = 1, b = 1, \alpha = 90$, where α is their enclosed angle, the hypotenuse has length $\sqrt{2}$. So what is $\sqrt{2}$?

Theorem 1. *There is no rational number x such that $x^2 = 2$. Equivalently, for an x satisfying $x^2 = 2$, x may not also satisfy $x = \frac{p}{q}$ with $p, q \in \mathbb{Z}$, $\gcd(p, q) = 1$*

Proof. Assume the statement is true. Then

$$\frac{p^2}{q^2} = 2 \implies p^2 = 2q^2 \implies 2 \mid p^2 \implies 2 \mid p \implies 4 \mid p^2 \implies \quad (1)$$

$$\implies 4 \mid 2q^2 \implies 2 \mid q^2 \implies 2 \mid q \implies \gcd(p, q) \geq 2 \nLeftarrow \quad (2)$$

Or alternatively, we could have also finished the proof via noticing that in the equation $p^2 = 2q^2$, p^2 has two in its prime factorization to an even power (since it is a square), while $2q^2$ has two in its prime factorization to an odd power (since it is a square, but multiplied by 2). Therefore our equation cannot hold, since one number cannot have two in its prime factorization to an odd and an even power at the same time. $\square$

We can generalize this result to show that for all $a \in \mathbb{Z}_{\geq 0}$, if $x^2 = a$, then if a contains a prime factor to an odd exponent (so a is not a perfect square), then and only then $x \in \mathbb{R} \setminus \mathbb{Q}$.

Remark 1 (Proof of irrationality of the square root of non-square numbers). *If $x^2 = a$ and a is not a perfect square, then and only then $x \in \mathbb{R} \setminus \mathbb{Q}$.*

Proof. If a is not a perfect square, then a contains a prime factor to an odd exponent. Then, if a were rational, we would have $a = \frac{p}{q}$ for some $p, q \in \mathbb{Z}$. (Note that, without loss of generality, we could also assume that $q > 0$ and that $\gcd(p, q) = 1$.)

$$\text{Then, since } a = \frac{p}{q}, \frac{p^2}{q^2} = a \implies p^2 = a \cdot q^2.$$

Now, pick a prime factor in a whose exponent is odd. (We know one must exist, since a is not a perfect square.)

- in the prime factorization of p , r must appear an even number of times, since it is a perfect square.
- in the prime factorization of aq^2 , r must appear an odd number of times, since the above argument applies to q^2 , and multiplying that once by a yields an odd exponent.

We have reached a contradiction, as no number can have a prime in its prime factorization to an even and an odd power at the same time, and prime factorizations are unique by the Fundamental Theorem of Arithmetic. Therefore $a \in \mathbb{R} \setminus \mathbb{Q}$. $\square$

2 Finding $\sqrt{2}$

Since we cannot find irrational numbers exactly, we will approximate it. Let us start with:

$$1 < x < 2 \Leftrightarrow 1 < x^2 < 4$$

Theorem 2.

$$0 < x < y \implies 0 < x^2 < y^2$$

Before we prove our theorem, let us write out the axioms of order (also known as axioms of arrangement):

1. Axiom of Trichotomy: Exactly one of the following can be true: $a < b$, $b < a$, $a = b$.
2. Addition preserves order: $a < b$ and c arbitrary $\implies a + c < b + c$.
3. Axiom of Transitivity: $a < b$ and $b < c \implies a < c$.
4. Multiplication by positive preserves order: $0 < a$ and $b < c$, then $ab < ac$.
5. Positivity of 1: $0 < 1$.

We will use the distributivity of addition (or that addition preserves order) to prove the following:

$$x < 0 \implies 0 < -x$$

Proof.

$$\begin{aligned} x + (-x) &< 0 + (-x) \\ 0 &< -x \end{aligned}$$

□

Now we use the distributivity of multiplication (or that multiplication by positive preserves order) to prove the following:

$$(-1) \cdot x = -x$$

Proof.

$$\begin{aligned} (1 + (-1)) \cdot x &= 0 \\ 1 \cdot x + (-1) \cdot x &= 0 \implies (-1) \cdot x = -x \end{aligned}$$

□

For this however, we need the following statement as well, for which we will again use the distributivity of multiplication:

$$0 \cdot x = 0$$

Proof.

$$\begin{aligned}(0 + 0) \cdot x &= 0 \cdot x \\ 0 \cdot x + 0 \cdot x &= 0 \cdot x \\ 0 \cdot x &= 0\end{aligned}$$

□

Now we proceed with a proof of Theorem 2 using the distributivity of multiplication:

Proof.

$$0 < x < y \implies 0 < x^2 < y^2$$

Then multiplying by x yields:

$$x^2 < xy$$

and multiplying the inequality by y gives us:

$$xy < y^2$$

Combining the latter two under the Axiom of Transitivity yields:

$$x^2 < xy < y^2 \implies x^2 < y^2$$

□

Remark 2. *We can also prove that $y < x < 0 \implies 0 < -x < -y$ using the distributivity of multiplication.*

Proof.

Assume

$$y < x < 0.$$

Multiplying the inequality by -1 reverses the order:

$$-y > -x > 0.$$

Rewriting in increasing order gives

$$0 < -x < -y.$$

□

3 Lion Catching Method

We can find $\sqrt{2}$ algorithmically via the **Bisection Method**, also known as the *Interval Halving Method*, or the *Lion Catching Method*. (from *Hector Petard: A Contribution to the Mathematical Theory of Big Game Hunting*, 1938.) Given $[a_0, b_0]$, $c_0 = \frac{a_0+b_0}{2}$ is the arithmetic mean, since

$$\begin{aligned}\frac{a+b}{2} - a &= b - \frac{a+b}{2} \\ \frac{b-a}{2} &= \frac{b-a}{2}\end{aligned}$$

In other words,

$$a + \frac{b-a}{2} = \frac{a+b}{2}.$$

Then we ask a simple question: $x < c_0$ (A) or $x > c_0$ (B)?

If (A), then $a_1 = a_0$, $b_1 = c_0$. Otherwise (B), $a_1 = c_0$, $b_1 = b_0$.

We will give the algorithm a limit, so that the program halts and we can retrieve the calculation. This limit will be its *precision*. (Alternatively, we could make the program report on its progress while storing the previous value in memory for a little longer, and continue [in theory] indefinitely.)

Theorem 3 (Cantor's Axiom). *Every nested sequence of non-empty closed intervals has at least one common point. Additionally, if the intervals' length goes to zero, then this yields exactly one common point.*

This axiom gives us a natural explanation of the Real Numbers ($\mathbb{R}$), since the axiom does not hold as long as we are in the set of rational numbers ($\mathbb{Q}$), but holds without exception in the set of real numbers.

Example: $I_n = [0, 1 + \frac{1}{n}]$ which are nested, non-empty closed intervals. Nested, since

$$1 + \frac{1}{n} > 1 + \frac{1}{n+1}.$$

Thus we get:

$$[0, 1] \subset I_n$$

Remark 3 (But what about open intervals?). *When we consider open intervals, like $I_n = (0, \frac{1}{n})$, there is no guarantee that the intersection is non-empty (as in our example).*

Now let's find a nested sequence of closed non-empty intervals whose intersection is *exactly* one point. The length of the intervals $\frac{I_0}{2^n} < \epsilon \implies$

4 Infinities

4.1 Problem 1.

There is an infinite amount of gold. A person takes one each day for infinitely many days. Can he take all of them?

Proposition 1. *He can't steal all of them, since any arbitrary day I can go in and observe that there is still an infinite amount of gold in the room.*

However, we will see that this is actually not the right way to look at the problem, since, as we will see, "any arbitrary day" is not equivalent to infinity. Instead, we observe the following proposition:

Proposition 2. *He can steal all of them, therefore there doesn't exist gold that he has not stolen.*

4.2 Bijection, Countably Infinite, Uncountably Infinite

For our proof, we will introduce concepts such as **bijection**, **countably infinite sets** and **uncountably infinite sets**.

Definition 1 (Countably infinite set). *A set is countably infinite if its elements can be put in one-to-one correspondence with the natural numbers.*

Definition 2 (Bijection). *A bijection is a function that is both injective and surjective. Intuitively, it pairs elements of one set with **exactly one unique element** of another set, and every element of the second set is matched to.*

Formally:

Let $f : A \rightarrow B$ be a function.

f is a bijection if it satisfies the following properties:

1. **Injective** (one-to-one)

$$f(a_1) = f(a_2) \implies a_1 = a_2$$

Different inputs always produce different outputs.

2. **Surjective** (onto)

$$\forall b \in B \exists a \in A : f(a) = b$$

Every element of B is the image of some element in A .

If both hold, then

$$f : A \leftrightarrow B$$

pairs the set perfectly. This means that the sets are the same size, or equivalently

$$|A| = |B|.$$

Theorem 4 (Inverse of a Bijection). *A function $f : A \rightarrow B$ is a bijection if and only if there exists a function $g : B \rightarrow A$ (often denoted as f^{-1}) such that:*

1. $g(f(a)) = a$ for all $a \in A$ (equivalently, $g \circ f = id_A$)
2. $f(g(b)) = b$ for all $b \in B$ (equivalently, $f \circ g = id_B$)

Proof. **Part 1: Forward Direction ($\implies$)**

Assume $f : A \rightarrow B$ is a bijection. We must show that an inverse function $g : B \rightarrow A$ exists.

- Because f is **surjective**, for every element $b \in B$, there is at least one element $a \in A$ such that $f(a) = b$.
- Because f is **injective**, this element a is strictly unique. There is only one a that maps to a specific b .
- Therefore, we can unambiguously define a new function $g : B \rightarrow A$ by setting $g(b) = a$.
- By this definition, applying f to $g(b)$ yields $f(g(b)) = f(a) = b$ for all $b \in B$.
- Similarly, applying g to $f(a)$ yields $g(f(a)) = g(b) = a$ for all $a \in A$.
- Thus, the inverse function g exists and satisfies both conditions.

Part 2: Reverse Direction ($\impliedby$)

Assume there exists a function $g : B \rightarrow A$ such that $g(f(a)) = a$ and $f(g(b)) = b$. We must show that f is a bijection (both injective and surjective).

- **Injective:** Suppose $f(a_1) = f(a_2)$ for some $a_1, a_2 \in A$. Applying our function g to both sides gives:

$$g(f(a_1)) = g(f(a_2))$$

Because we know $g(f(a)) = a$, this equation simplifies directly to:

$$a_1 = a_2$$

Since $f(a_1) = f(a_2) \implies a_1 = a_2$, the function f is injective.

- **Surjective:** Let b be an arbitrary element in B . We need to find an element $a \in A$ such that $f(a) = b$. Let $a = g(b)$. Applying f to this chosen a gives:

$$f(a) = f(g(b))$$

Because we know $f(g(b)) = b$, we have:

$$f(a) = b$$

Therefore, for every $b \in B$, there is an $a \in A$ that maps to it, making f surjective.

Since f is proven to be both injective and surjective, it is, by definition, a bijection. $\square$

With our newfound knowledge, let us revisit *countably infinite sets*.
Formally, a set A is countably infinite if there exists a **bijection**

$$f : \mathbb{N} \rightarrow A.$$

Such sets include:

- $\mathbb{N}$
- $\mathbb{Z}$
- $\mathbb{Q}$.

This means that you can list the elements

$$a_1, a_2, a_3, \dots$$

even though the list never ends.

Definition 3 (Uncountably infinite set). *A set is uncountable if there is no enumeration*

$$r_1, r_2, r_3, \dots$$

that contains every element of the set. The classic example is $\mathbb{R}$, proved by Cantor's diagonalization argument.

With these concepts in mind now, let us reconsider the original gold problem.

Proof. Since the set of all gold bars is $\mathbb{N}$ (as they can be organised next to each other, like listing them), and the set of all days is also $\mathbb{N}$ since we can also "list them out", both sets are countably infinite, and thus their cardinalities are equal. This tells us that the vault *can be exhausted*. (Note that this proof is not "rigorous" in any mathematical sense, as the question is also not posed in a mathematical sense. Refer to *Cantor's diagonalization argument*.) $\square$

5 Factorizing expressions with square roots

Let's try to expand $(\sqrt{2} + 1)^2$.

$$(\sqrt{2} + 1)^2 = (\sqrt{2})^2 + 2\sqrt{2} + 1 = 3 + 2\sqrt{2}$$

Now let us try to factorize $11 + 6\sqrt{2}$. What would be a more systematic approach? We first observe that our solution will be of the form

$$(a + b)^2 = a^2 + b^2 + 2ab,$$

(Trivially, since our equation is of the form $a^2 + b^2 + 2ab$.)

Then we notice that the $2ab$ part will be:

$$6\sqrt{2} = 2ab$$

And that

$$3\sqrt{2} = ab.$$

Which gives us a table of possible (a, b) solutions.

a	b
1	$3\sqrt{2}$
3	$\sqrt{2}$

We can see that we need $a^2 + b^2 = 11$.

$$(1)^2 + (3\sqrt{2})^2 = 19,$$

Therefore the solution must be the other case, $(a = 3, b = \sqrt{2})$. Verifying,

$$(3)^2 + (\sqrt{2})^2 = 9 + 2 = 11.$$

Which means

$$11 + 6\sqrt{2} = (3 + \sqrt{2})^2$$

□

Now let's try a little more complicated problem. Factorize $17 - 4\sqrt{15}$. Recall that

$$(a + b)^2 = a^2 + b^2 + 2ab$$

And that

$$(a - b)^2 = a^2 + b^2 - 2ab$$

Therefore

$$2ab = 4\sqrt{15}$$

$$ab = 2\sqrt{15}$$

Now we have the following table of pairs of possible solutions (a, b) which satisfy $ab = 2\sqrt{15}$.

a	b
2	$\sqrt{15}$
1	$2\sqrt{15}$
$\sqrt{3}$	$2\sqrt{5}$
$\sqrt{5}$	$2\sqrt{3}$

Evaluating all the possible solution pairs to find one which also satisfies $a^2 + b^2 = 17$:

Case 1.

$$(2)^2 + (\sqrt{15})^2 = 19$$

does not satisfy. Case 2:

$$(1)^2 + (2\sqrt{15})^2 = 61$$

also does not satisfy our equation. Case 3:

$$(\sqrt{3})^2 + (2\sqrt{5})^2 = 23$$

again, does not satisfy $a^2 + b^2 = 17$. Therefore, case 4;

$$(\sqrt{5})^2 + (2\sqrt{3})^2 = 17.$$

Which means that $a = \sqrt{5}$, $b = (2\sqrt{3})$.

Thus

$$17 - 4\sqrt{15} = (\sqrt{5} - 2\sqrt{3})^2$$

□

Now another: Factorize $120 - 30\sqrt{15}$. We notice that the solution will be of form $(a - b)^2 = a^2 + b^2 - 2ab$. Then

$$2ab = 30\sqrt{15}$$

$$ab = 15\sqrt{15}$$

Which gives us the following table of possible solution pairs (a, b):

a	b
1	$15\sqrt{15}$
3	$5\sqrt{15}$
5	$3\sqrt{15}$
$3\sqrt{5}$	$5\sqrt{3}$
$5\sqrt{3}$	$3\sqrt{5}$

We find, after exhaustive enumeration, that only the pair $(3\sqrt{5}, 5\sqrt{3})$ satisfies $a^2 + b^2 = 120$, and therefore

$$120 - 30\sqrt{15} = (3\sqrt{5} - 5\sqrt{3})^2$$

□

6 Algebra with Square Roots: Identities, Definition

Definition 4 (Principal Square Root). *We denote the principal square root via the symbol $\sqrt{\cdot}$. To ensure this operation is a well-defined function, we restrict its codomain to non-negative real numbers. Thus,*

$$\sqrt{x} = a \iff a^2 = x$$

where $a \in \mathbb{R}$ and $a \geq 0$.

Remark 4 (Why not "Without Loss of Generality"?). *We cannot state $a \geq 0$ "without loss of generality." The equation $a^2 = x$ (for $x > 0$) has two distinct real solutions. If we assumed $a \geq 0$ without loss of generality, it would imply that choosing $a \leq 0$ yields the exact same mathematical properties. Instead, restricting $a \geq 0$ is a strict definitional choice required to make the square root a single-valued function (specifically, the **principal** branch).*

6.1 Identities

Identity 1.

$$(\sqrt{a})^2 = a.$$

Identity 2.

$$\sqrt{a^2} = |a|.$$

Identity 3.

$$\sqrt{a} \cdot \sqrt{b} = \sqrt{ab}.$$

Where $a, b \geq 0$.

Identity 4.

$$\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}.$$

Where $a, b \geq 0$. One common mistake is thinking that

$$\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}$$

is an identity. While in reality, it is not. For example,

$$\sqrt{16+9} \neq \sqrt{16} + \sqrt{9}$$

since

$$5 \neq 4 + 3.$$

The statement $\sqrt{ab} = \sqrt{a} \cdot \sqrt{b}$ is not equivalent to the statement $\sqrt{a} \cdot \sqrt{b} = \sqrt{ab}$, since the existence of $\sqrt{ab}$ does not guarantee the existence of $\sqrt{a}$ or $\sqrt{b}$, unless the condition $a, b \geq 0$ is given.

$$\sqrt{20} = \sqrt{4 \cdot 5} = \sqrt{4} \cdot \sqrt{5} = 2\sqrt{5}.$$

$$3\sqrt{2} = \sqrt{9} \cdot \sqrt{2} = \sqrt{18}.$$

Generally, $a\sqrt{b} = \sqrt{a^2b}$. and $\sqrt{c} = m\sqrt{n}$ where $c = m^2n$. (Every number is of this form, however, m may be 1 for some cases, therefore this may not lead to an actual reduction.)

How is this useful? We may ask, which of the following is bigger?

$$4\sqrt{2}$$

6.2 Problems

6.2.1 Evaluating expressions

1. $\sqrt{12} - \sqrt{27} + \frac{1}{2}\sqrt{48}$;
2. $3\sqrt{2} + \sqrt{32} - \sqrt{200}$;
3. $2\sqrt{72} - \sqrt{50} - 2\sqrt{8}$;
4. $5\sqrt{3} + \frac{1}{3}\sqrt{27} - \sqrt{48}$;
5. $\sqrt{80} + \frac{1}{2}\sqrt{20} + 3\sqrt{45}$;
6. $3\sqrt{32} - \sqrt{50} + 2\sqrt{18} + 3\sqrt{8}$;
- 7.
- 8.

6.2.2 Solutions.

1. $2\sqrt{3} - 3\sqrt{3} + 2\sqrt{3} = \sqrt{3}$
- 2.
3. $12\sqrt{2} - 5\sqrt{2} - 4\sqrt{2}$

7 Rationalization

Definition 5 (Rationalization). *Rationalization is the process of eliminating a square root (or other **radical**) from the denominator of a fraction by multiplying by a suitable expression. Verb: Rationalizing [the denominator].*

Definition 6 (Radical). *A radical expression is an expression of the form $\sqrt[n]{x}$.*

Definition 7 (Rational). *A rational number is any number of the form $\frac{p}{q}$ where $p, q \in \mathbb{Z}$ and $q \neq 0$. (The set of rational numbers is $\mathbb{Q}$).*

Example 1.

$$\frac{8}{\sqrt{2}} = \frac{8\sqrt{2}}{\sqrt{2}\sqrt{2}} = \frac{8\sqrt{2}}{2} = 4\sqrt{2}$$

Example 2.

$$\frac{\sqrt{5}}{\sqrt{7}} = \frac{\sqrt{5}\sqrt{7}}{7} = \frac{\sqrt{35}}{7}$$

We can see that the fraction does not necessarily disappear.

Example 3.

$$\frac{2}{2 + \sqrt{2}}.$$

7.1 Problems

$$\frac{1}{\sqrt{2} + \sqrt{3} - \sqrt{5}}$$

Multiplying by

$$\frac{\sqrt{2} - \sqrt{3} + \sqrt{5}}{\sqrt{2} - \sqrt{3} + \sqrt{5}}$$

7.2 Hard problems

7.2.1 Problem 1.

7.2.1.1

Problem

Simplify the following expression:

$$\frac{2 - \sqrt{3}}{\sqrt{2} - \sqrt{2 - \sqrt{3}}} + \frac{2 + \sqrt{3}}{\sqrt{2} + \sqrt{2 + \sqrt{3}}}.$$

7.2.1.2

Solution

$$\begin{aligned} \frac{(2 - \sqrt{3})(\sqrt{2} + \sqrt{2 - \sqrt{3}})}{2 - (2 - \sqrt{3})} &= \frac{(2 - \sqrt{3})(\sqrt{2} + \sqrt{2 - \sqrt{3}})}{\sqrt{3}} \\ \frac{(2 + \sqrt{3})(\sqrt{2} - \sqrt{2 + \sqrt{3}})}{2 - (2 + \sqrt{3})} &= -\frac{(2 + \sqrt{3})(\sqrt{2} - \sqrt{2 + \sqrt{3}})}{\sqrt{3}} \\ \frac{(2 - \sqrt{3})(\sqrt{2} + \sqrt{2 - \sqrt{3}}) - (2 + \sqrt{3})(\sqrt{2} - \sqrt{2 + \sqrt{3}})}{\sqrt{3}} \\ \frac{2\sqrt{2} - \sqrt{6} + 2\sqrt{2 - \sqrt{3}} - \sqrt{3}\sqrt{2 - \sqrt{3}} - 2\sqrt{2} - \sqrt{6} + 2\sqrt{2 + \sqrt{3}} + \sqrt{3}\sqrt{2 + \sqrt{3}}}{\sqrt{3}} \end{aligned}$$

We can see that the $2\sqrt{2}$ s cancel out, and grouping like terms gives us

$$\frac{-2\sqrt{6} + 2(\sqrt{2 - \sqrt{3}} + \sqrt{2 + \sqrt{3}}) + \sqrt{3}(\sqrt{2 + \sqrt{3}} - \sqrt{2 - \sqrt{3}})}{\sqrt{3}}.$$

We now observe the following particular parts:

$$\text{A) } \sqrt{2 - \sqrt{3}} + \sqrt{2 + \sqrt{3}}$$

$$\text{B) } \sqrt{2 + \sqrt{3}} - \sqrt{2 - \sqrt{3}}$$

Idea for A): Let's square the expression, then see if it simplifies, then take its square root!

$$\sqrt{2 - \sqrt{3}} + \sqrt{2 + \sqrt{3}}$$

$$\frac{\sqrt{(\sqrt{2+\sqrt{3}}+\sqrt{2-\sqrt{3}})^2}}{\sqrt{2+\sqrt{3}+2-\sqrt{3}+2\sqrt{4-3}}}$$

$$\sqrt{6}.$$

The idea is the same for B):

$$\frac{\sqrt{(\sqrt{2+\sqrt{3}}-\sqrt{2-\sqrt{3}})^2}}{\sqrt{2+\sqrt{3}+2-\sqrt{3}-2\sqrt{4-3}}}$$

$$\sqrt{2}.$$

Which lets us finish our solution:

$$\frac{-2\sqrt{6}+2\sqrt{6}+\sqrt{3}\sqrt{2}}{\sqrt{3}}$$

$$\boxed{\sqrt{2}}.$$

□

7.2.1.2.1

An alternate approach

What if $\sqrt{2+\sqrt{3}}$ is a square under a square root?

$$\sqrt{2+\sqrt{3}} = \sqrt{\frac{4+2\sqrt{3}}{2}} = \sqrt{\frac{(\sqrt{3}+1)^2}{2}} = \frac{\sqrt{3}+1}{\sqrt{2}}$$

Similarly for $\sqrt{2-\sqrt{3}}$:

$$\sqrt{2-\sqrt{3}} = \sqrt{\frac{4-2\sqrt{3}}{2}} = \sqrt{\frac{(\sqrt{3}-1)^2}{2}} = \frac{\sqrt{3}-1}{\sqrt{2}}$$

Or collectively

$$\sqrt{2\pm\sqrt{3}} = \sqrt{\frac{4\pm 2\sqrt{3}}{2}} = \sqrt{\frac{(\sqrt{3}\pm 1)^2}{2}} = \frac{\sqrt{3}\pm 1}{\sqrt{2}}$$

Therefore

$$\sqrt{2+\sqrt{3}}+\sqrt{2-\sqrt{3}} = \frac{\sqrt{3}+1+\sqrt{3}-1}{\sqrt{2}} = \frac{2\sqrt{3}}{\sqrt{2}} = \sqrt{2}\sqrt{3} = \sqrt{6}$$

And

$$\sqrt{2+\sqrt{3}} - \sqrt{2-\sqrt{3}} = \frac{\sqrt{3}+1 - \sqrt{3}+1}{\sqrt{2}} = \frac{2}{\sqrt{2}} = \sqrt{2}.$$

From here, we can finish our solution in exactly the same way as the previous approach:

$$\frac{-2\sqrt{6} + 2\sqrt{6} + \sqrt{3}\sqrt{2}}{\sqrt{3}}$$

$$\boxed{\sqrt{2}.}$$

□

7.2.2

Summary

Summarizing, we first asked to simplify the following expression:

$$\frac{2-\sqrt{3}}{\sqrt{2}-\sqrt{2-\sqrt{3}}} + \frac{2+\sqrt{3}}{\sqrt{2}+\sqrt{2+\sqrt{3}}}.$$

After some simplification, we arrived at

$$\frac{-2\sqrt{6} + 2(\sqrt{2-\sqrt{3}} + \sqrt{2+\sqrt{3}}) + \sqrt{3}(\sqrt{2+\sqrt{3}} - \sqrt{2-\sqrt{3}})}{\sqrt{3}},$$

and then we asked how the expression $\sqrt{2 \pm \sqrt{3}} \pm \sqrt{2 \pm \sqrt{3}}$ can be simplified with the restriction that the two outermost added terms have alternating signs. We proposed two approaches: In the first, we tried the first thing that comes to mind: square, then try to simplify, because maybe the numbers play out nicely, then take its square root. In the second, we noticed that both terms in the addition are actually squares under a square root, and simplified using this fact.

Our result was $\sqrt{4 \pm 2}$, giving us the final solution of $\sqrt{2}$.

8 Exam Problems

8.1 Problem 1.

Evaluate the following: (3 points)

$$\sqrt{\frac{36}{25} \cdot \frac{81}{144} \cdot 0,36}$$

First, let $0,36 = \frac{36}{100}$, then split the square root into a product of fractions

$$= \sqrt{\frac{36}{25}} \cdot \sqrt{\frac{81}{144}} \cdot \sqrt{\frac{36}{100}} = \frac{6}{5} \cdot \frac{9}{12} \cdot \frac{6}{10} = \frac{6 \cdot 9 \cdot 6}{5 \cdot 12 \cdot 10} = \frac{324}{600} = \frac{27}{50} = 0,54$$

$$\sqrt{72 \cdot 32}$$

$$\sqrt{2^8 \cdot 3^2} = 2^4 \cdot 3 = 16 \cdot 3 = 48.$$

Generally,

$$\sqrt{ab}$$

Eval. ab 's prime factorization

$$ab = p_1^{\alpha_1} \cdot p_2^{\alpha_2} \dots p_i^{\alpha_i}$$

If all α_i are even, then

$$\sqrt{ab} = p_1^{\frac{\alpha_1}{2}} \cdot p_2^{\frac{\alpha_2}{2}} \dots p_i^{\frac{\alpha_i}{2}}$$

$$\sqrt{90 \cdot 64}$$

$$\sqrt{90 \cdot \frac{64}{10}}$$

$$\sqrt{10 \cdot 9 \cdot \frac{64}{10}}$$

$$\sqrt{9 \cdot 64}$$

$$\sqrt{9} \cdot \sqrt{64} = 3 \cdot 8 = 24.$$

8.2 Problem 2.

Evaluate the following expressions (2 points)

$$2\sqrt{72} - \sqrt{50} - 2\sqrt{8}$$

$$\sqrt{72} = \sqrt{36 \cdot 2} = 6\sqrt{2}$$

$$\sqrt{50} = \sqrt{25 \cdot 2} = 5\sqrt{2}$$

$$\sqrt{8} = \sqrt{4 \cdot 2} = 2\sqrt{2}$$

$$2(6\sqrt{2}) - 5\sqrt{2} - 2(2\sqrt{2}) = 12\sqrt{2} - 5\sqrt{2} - 4\sqrt{2} = 3\sqrt{2}$$

$$\sqrt{12} - \sqrt{27} + \frac{1}{2}\sqrt{48}$$

We notice that all of the numbers are divisible by 3.

$$2\sqrt{3} - 3\sqrt{3} + \frac{1}{2}4\sqrt{3}$$

$$2\sqrt{3} - 3\sqrt{3} + 2\sqrt{3}$$

$$\sqrt{3}.$$

8.3 Problem 3.

Simplify the following expressions. (2 points)

$$\begin{aligned}\sqrt{30} \cdot \sqrt{20} \\ \sqrt{30 \cdot 20} &= \sqrt{600} \\ 600 &= 2^3 \cdot 3 \cdot 5^2 \\ &= \sqrt{2^2 \cdot 5^2 \cdot 2 \cdot 3} = 2 \cdot 5 \cdot \sqrt{6} = 10\sqrt{6}\end{aligned}$$

$$\begin{aligned}\sqrt{63} \cdot \sqrt{28} \\ \sqrt{1764} &=? \\ 1764 &= 2^2 \cdot 3^2 \cdot 7^2 \\ \sqrt{2^2 \cdot 3^2 \cdot 7^2} &= 2 \cdot 3 \cdot 7 = 42.\end{aligned}$$

8.4 Problem 4.

Evaluate the expression. (3 points)

$$(2\sqrt{18} + \sqrt{54}) \cdot (3\sqrt{6} - \sqrt{27} - \sqrt{3})$$

Simplify every radical

$$\begin{aligned}2\sqrt{18} &= 2 \cdot 3\sqrt{2} = 6\sqrt{2} \\ \sqrt{54} &= 3\sqrt{6} \\ \sqrt{27} &= 3\sqrt{3}\end{aligned}$$

Then the expression becomes

$$\begin{aligned}(6\sqrt{2} + 3\sqrt{6}) \cdot (3\sqrt{6} - 3\sqrt{3} - \sqrt{3}) \\ (6\sqrt{2} + 3\sqrt{6}) \cdot (3\sqrt{6} - 4\sqrt{3})\end{aligned}$$

After simply algebraically expanding the brackets like with any expression, we get

$$54 + 36\sqrt{3} - 36\sqrt{2} - 24\sqrt{6}.$$

8.5 Problem 5.

Calculate the exact value of the following expression. (3 points)

$$(\sqrt{2 + 2\sqrt{3}} - \sqrt{-2 + 2\sqrt{3}})^2$$

We will omit the details here, ask any LLM. The final result, for checking purposes, is $4(\sqrt{3} - \sqrt{2})$.

8.6 Problem 6.

Calculate the exact value of the following expression. (3 points)

$$\sqrt{48 + 6\sqrt{15}} + \sqrt{48 - 6\sqrt{15}}$$

We will omit the details here, ask any LLM. The final results, for checking purposes, is $6\sqrt{5}$.

8.7 Problem 7.

Rationalize the denominator.

$$\frac{2}{\sqrt{5} - 2}$$

The conjugate of $\sqrt{5} - 2$ is $\sqrt{5} + 2$, therefore we will multiply the entire fraction by that.

$$\frac{2(\sqrt{5} + 2)}{(\sqrt{5} - 2)(\sqrt{5} + 2)}$$
$$2\sqrt{5} + 4.$$

$$\frac{28}{1 + \sqrt{5}}$$

Multiply by the conjugate

$$\frac{28(1 - \sqrt{5})}{(1 + \sqrt{5})(1 - \sqrt{5})}$$

We omit the small details. The final result is

$$7(\sqrt{5} - 1).$$

8.8 Problem 8.

Simplify the following expression. (3 points)

$$\sqrt{a + 2 + 4\sqrt{a - 2}}$$